

Hurricane Disruption by Electrical-conversion System

(H.A.D.E.S)

Background

Hurricanes are a destructive force of nature, capable of wreaking great havoc when they make landfall. In America, the damage caused is estimated to be about \$28 billion a year [1].

Hurricane Katrina alone caused \$161 billion in damages when she struck in 2005 [2]. It is a problem so severe to the point where the US President Donald Trump recently suggested bombing hurricanes with nuclear weapons to disrupt them before they made landfall [3]. Not only would it not work, the resulting detonation would release tons of radioactive nuclear fallout that can be spread far and wide by the high speed winds of the hurricane.

Hurricanes form when masses of cold and warm air collide. There have been many unsuccessful suggestions on how to prevent their formation, primarily by lowering the temperature of the sea surface, such as by towing icebergs, seeding clouds with sea salt and using aircraft to release dry ice in the vicinity of hurricanes [4]. However, instead of waiting for a way to completely prevent the formation of hurricanes, renowned astrophysicist Neil deGrasse Tyson, in an interview on the Today Show, suggested looking into harnessing the power of hurricanes to generate electricity. [5]

Problem

Currently in the market, there are a few products that convert energy from storm winds to electricity. However, they are stationary windmills that are either mounted on the roof [6] or anchored to the seabed [7]. These designs are primarily used for generating electricity, and may not be effective for disrupting a hurricane because once the hurricane moves past them, they will stop converting the hurricane's energy to electricity, leaving the hurricane with just a minor loss of energy. The purpose of this proposal is to create a machine that disrupts hurricane activity by continuously converting the energy of the hurricane into electrical energy.

Solution

From the Law of Conservation of Energy¹, we know that energy cannot be created or destroyed; it can only be converted from one form to another. In other words, all the energy that the hurricane possesses has to come from somewhere, and by tapping into it, we can convert that latent energy into electricity instead of allowing the hurricane to convert it into mechanical energy in the form of strong winds.

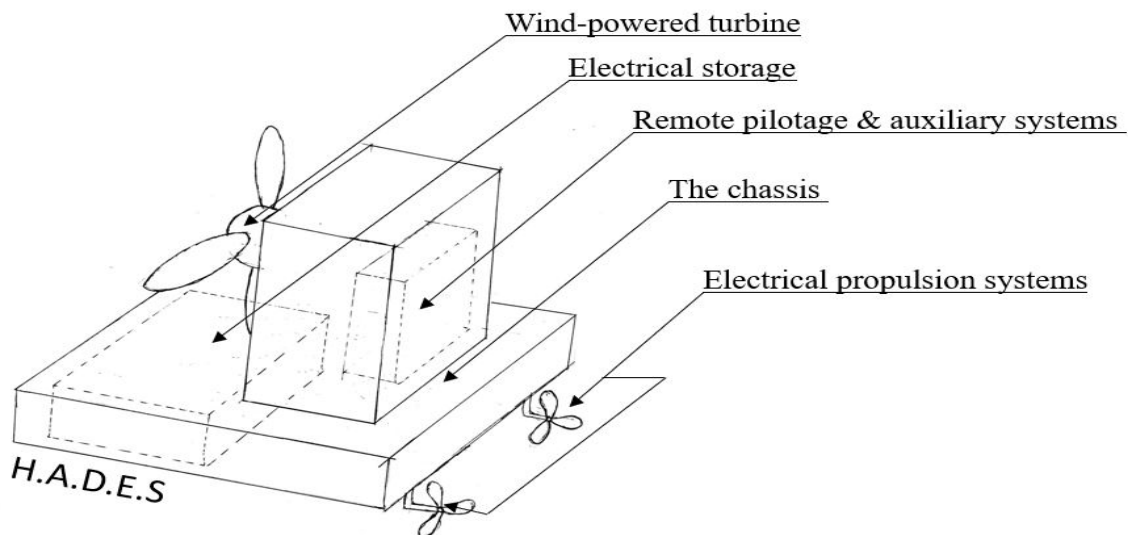
In theory, once we have fully converted all that energy into electricity, the hurricane would have been prevented from forming, or would cease to exist if it has already been created. *H.A.D.E.S* is basically a mobile windmill that will follow the course of the hurricane and convert the hurricane's force into electricity.

The whole system is made up of 5 main systems:

¹ The Law of Conservation of Energy states that the total energy of an isolated system remains constant; it is said to be conserved over time.[8]

- The chassis - which all the subsystems will be mounted.
- Electrical propulsion systems - enables *H.A.D.E.S* to follow the hurricane's path.
- Wind-powered turbine - converts the hurricane's power to electricity.
- Electrical storage - stores the converted energy for the propulsion and auxiliary systems.
- Remote pilotage and auxiliary systems - enables *H.A.D.E.S* to be remotely piloted, and for various systems required for marine vessels.

Figure 1:



Because hurricanes form over the ocean, this windmill will be mounted on a barge which will be remotely piloted by a specialized ship. The barge will be towed to a safe area outside the hurricane, after which the crew will remotely control it to enter the hurricane and start the energy conversion process. As there is a limit to how much electricity can be stored, once electrical storage is full, the remaining electricity will be discharged into the sea.

Figure 2:

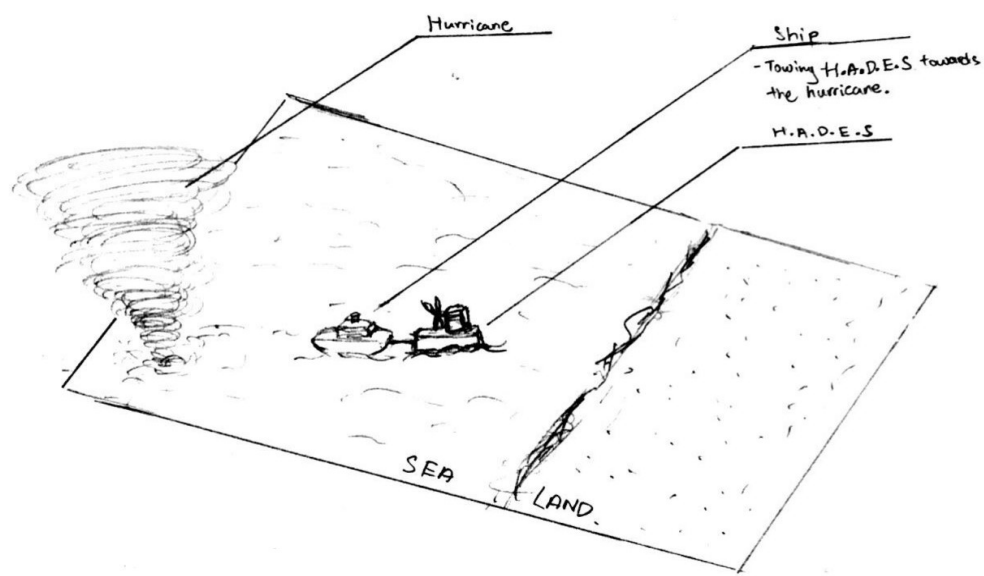
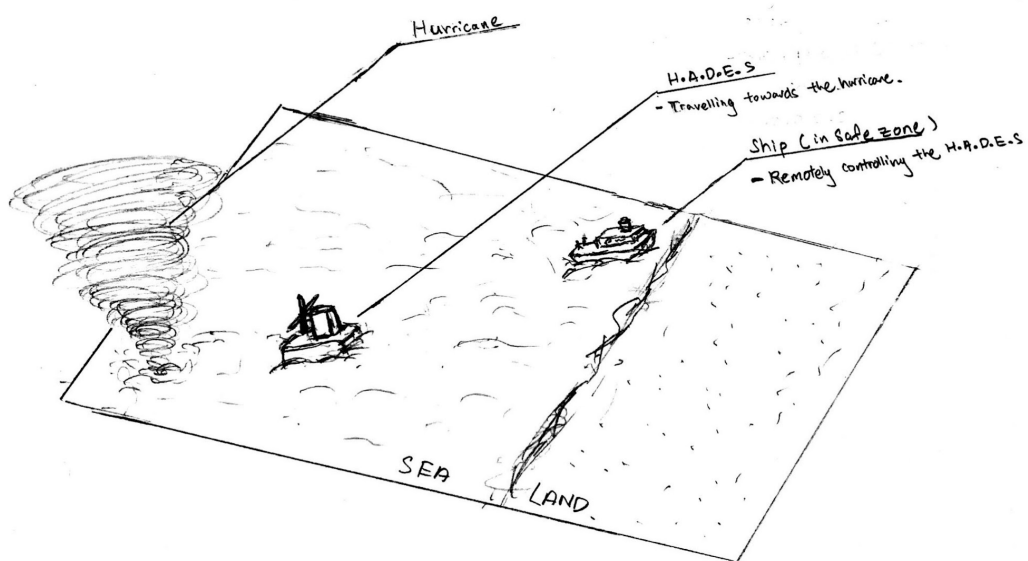


Figure 3:



Benefits

H.A.D.E.S. will benefit countries that experience typhoons or hurricanes frequently. Countries which purchase this product will spend less or even nothing in relief and reconstruction efforts.

The money saved can be put to better use, such as for research into climate change. People's lives will not be lost or affected by the destruction that would otherwise befall the area where a hurricane might strike, leading to minimal downturn in the economy.

Implementation

We will require the following in order to produce this product:

- Collaboration with ST Marine to design and build the barge
- Collaboration with Tesla to design the windmill and electricity storage and discharge
- Work with Oceanvolt, a company specializing in electric marine propulsion systems

Cost

R&D:	\$3 million
Windmill:	\$3 million
High capacity batteries:	\$2 million
Remote control systems:	\$500 000
Propulsion systems:	\$1 million

Construction:	\$4 million
Total prototype cost:	\$13.5 million

Conclusion

Because of climate change and global warming, there has been an increase in hurricane activity [9]. *H.A.D.E.S* is an immediate solution to prevent or reduce hurricane damage so that funds used for relief efforts and reconstruction can be used elsewhere. With *H.A.D.E.S*, we are a step closer to reducing hurricane activity, and ultimately, a better Earth for everyone.

References

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